# SILICON DREAMS

*Module 1 · Study Guide*

**MODULE 1 · WEEK 1 · CM-HW-101-M1**

**SG-M1-01 · paired with TB-M1-01**

## Platform Setup Checklist

*GitHub, ChipFoundry project creation, toolchain install, smoke-test run.*

### Goal

Before you write any Verilog, you need a working environment end to end. This guide is a checklist — do not read it in a hurry. Every step is something that will silently fail later if you skip it.

### Part A — Account and repository

- ☐ Have a GitHub account with a verified email.
- ☐ Fork the starter repo: [https://github.com/jdicorpo/new-fpga-verilog-test](https://github.com/jdicorpo/new-fpga-verilog-test) → 'Use this template' → new repo under your own username.
- ☐ Clone your fork locally to a path with no spaces in it.
- ☐ Create a ChipFoundry account at the course-provided portal URL and link it to your GitHub identity.

### Part B — Toolchain install

The toolchain installation can be done using the below command.
#Ubuntu 22.04 / Debian

- ***sudo apt-get update***
- ***sudo apt-get install -y git python3 python3-pip python3-venv iverilog gtkwave make build-essential***

# macOS (with Homebrew)

- ***brew install git python icarus-verilog gtkwave make***

# Create a virtual environment (recommended)

- ***python3 -m venv ~/.venvs/silicondreams***
- ***source ~/.venvs/silicondreams/bin/activate***

# Inside the virtualenv, install cocotb

- ***pip install -r <repo-path>/test/requirements.txt***

PS: Windows users should install Ubuntu WSL and follow the Ubuntu 22.04/Debian steps

Verify the installation

- ☐ Verify iverilog -V prints a version string.
- ☐ Verify gtkwave --version prints a version string.
- ☐ Verify python3 -c 'import cocotb' runs silently with exit code 0.

## Part C — Smoke-test the starter

- ***cd <repo-path>/test***
- ***Make or make smoke***

The expected output ends with 'PASS' in green and creates a tb.vcd file. Open it:

***gtkwave tb.vcd &***

- ☐ gtkwave opens a window. You see the clk signal toggling.
- ☐ Add ui_in, rst_n, and uo_out to the waveform pane. Confirm uo_out = 0xBF at the end of the run.

## Self-check rubric

- All Part A boxes ticked.
- All Part B boxes ticked.
- tb.vcd exists and opens in gtkwave.
- You can reproduce make → PASS in a single command.

Claim the 200 XP for module setup. Move to SG-M1-02.